

## Sustainability Analysis of Requirements Specification Checklist

Version

Reviewer(s)

Review date

### 1. Purpose

- Is the scope of project/software well defined?
- Are the problems clearly stated?
- Does the goals and objectives aligned with the scope and problems?

### 2. Framework for Sustainability of Software System Design (FSSSD)

| Question                                             | SDLC Phases                             | Sustainability Goals                                             | Sustainability Concepts, Methods and Tools                                                 | Indicators /Measure / Metric                                                            |
|------------------------------------------------------|-----------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Are the project definition clear and understandable? | Phase 1. Project Definition             | Does the sustainability align with the project definition?       | Was the project definition supported by any concepts of sustainability or tools or method? | Does the indicators relate to the project definition?                                   |
| Are the user requirements well defined?              | Phase 2. User Requirements Definition   | Are the sustainability goals inline with the user requirements?  | was the sustainability requirement template used?                                          | are there user requirements related to sustainability concerns?                         |
| Are the system requirement properly stated?          | Phase 3. System Requirements Definition | Does the sustainability goals align with the system requirements | Was goal model applied?                                                                    | Are there sustainability goals that relates to the different sustainability dimensions? |

### 3. Goal Model

- Are the goals clearly stated?
- Are the goals related to any of the sustainability dimensions?
- Are the goals measurable?
- Does the goals help identify conflicts between other goals (optional)?
- Does the goals model the system from business, usage and system categorization?
- Can you identify all business- relevant (strategic) goals as well as goals with direct impact on the system or project from business goals?
- Can you identify functional context and usage of the system (user perspective) from usage goals?
- Can you identify the target system characteristics from system goals?

### 4. Sustainability Requirement Template

- Are the requirements clearly stated?
- Are the requirements clearly categorized to the five sustainability dimensions?
- Are the explanation clear and understandable?
- Does the template provide better understanding of how the requirements relates to each of the sustainability dimensions?

## 5. Sustainability analysis Radar chart

- Are the first, second and third order impacts clearly stated?
- Does each of the five dimensions of sustainability clearly cover:
  - First order impact :(immediate effects) are about the direct effects of the development and use of a software system.
  - Second order impacts :(enabling effects) are about the indirect impacts related to the effects of using the software system in its application domain.
  - Third order impacts : (structural effects) are the cumulative long term effects resulting from accumulating first and second order impacts over time
- Does the sustainability analysis radar chart show the impact or potential impacts both positive and negative of the proposed solutions (software system)?
- Can you identify how to improve the proposed solution (software system) from the radar chart diagram?